

Unified Gateway Implementation Handbook

HF-managed evidence and policy plane

ARCHITECTURE

HF-MANAGED

EVIDENCE-FIRST

PRODUCTION GATES

Authoritative decision. VeriTrust executes no learned production inference in browser code, Vercel functions, platform workers, local runtimes, or VeriTrust-owned model servers. Learned production inference uses Hugging Face-managed infrastructure. The canonical hierarchy is HF-operated hf-inference when the exact model/task and all gates pass; then Hugging Face Inference Endpoint; then private Hugging Face repository plus Hugging Face Inference Endpoint. Third-party Hugging Face Inference Providers are DISABLED_BY_DEFAULT, outside the authoritative production path, and never an automatic fallback.

Status: implementation architecture and research package; no production performance claim. All UNKNOWN values remain explicit.

Prepared from the executable repository, seven historical PDFs, fresh official Hugging Face research, primary competitor sources and C2PA standards.

Contents and control

This is an Implementation Architecture Handbook, not a standalone executable specification. Critical invariants, model identities, deployment classes, evidence semantics, receipt fields, lifecycle states and companion references are generated from validated canonical structured sources; all provider availability is point-in-time and promotion-gated.

NORMATIVE IMPLEMENTATION COMPANIONS: `Architecture_Source_of_Truth.yaml`; `Model_Registry_Proposal.json`; `HuggingFace_Deployment_Matrix.csv`; `Implementation_Backlog.csv`; `Database_Target_Contract.md`; `Verification_Test_Plans.md`; `Migration_Roadmap.md`.

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Cross-document invariants

- HF-managed learned inference only; no normal third-party provider tier.
- Registry resolution precedes dispatch.
- Receipt identifies exact deployment, bundle and ordered learned components.
- `raw_output`, `raw_label`, `raw_score`, `normalized_label`, `normalized_score`, `calibrated_probability` and `calibration_version` remain distinct.
- Expected-evidence lifecycle uses the canonical 16-state enum: `planned`, `dispatched`, `running`, `completed`, `abstained`, `not_applicable`, `unavailable`, `timed_out`, `failed`, `cancelled`, `privacy_blocked`, `policy_skipped`, `insufficient`, `unsupported`, `degraded`, `budget_exceeded`.
- Specialist evidence cannot silently substitute or own final policy.
- No original PDF or production source file is overwritten.

1. Executive decision and current implementation

Registry-before-dispatch is the first architectural milestone. A receipt must prove the selected deployment actually ran.

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Current Gateway validates/authenticates, normalizes artifacts, selects policy, invokes adapters, persists evidence, correlates and reports. It has strong HMAC, idempotency, signed upload, SSRF, encrypted webhook and signed dispatch controls. The critical flaw is that model-version persistence resolves after inference.

2. Shared vocabulary and non-negotiable boundaries

Artifact is immutable content identity; evidence is a specialist observation; receipt is execution provenance; sufficiency says whether evidence can support a decision; policy alone authorizes action.

Specialist	Owns	Never owns
Phishing	email/text/auth/identity evidence	final allow/block
Link	URL/static/enrichment evidence	final allow/block
Deepfake	media role evidence	identity/payment action
Gateway	routing, graph, sufficiency, correlation, policy	learned model execution

3. Target control and data planes

The public API/control plane validates and plans. Durable queues carry resolution IDs and object references. Workers orchestrate remote HF calls. HF runs every learned model. Postgres/object storage retain immutable evidence and artifacts. No learned weights enter a Gateway worker.

CLIENT -> VERITRUST INGRESS -> IMMUTABLE ARTIFACT -> DETERMINISTIC FEATURES
-> PERSIST EXACT REGISTRY RESOLUTION -> HF-MANAGED INFERENCE
-> VALIDATE RESPONSE + RECEIPT -> SPECIALIST EVIDENCE
-> SUFFICIENCY + CORRELATION -> VERSIONED POLICY -> DECISION / HUMAN REVIEW

4. Artifact graph

Every submission creates one immutable root and typed children for MIME parts, extracted URLs, redirects, images/screenshots and derived crops. Each edge records extractor/version/reason. Derived artifacts reference parent hash; raw originals remain preserved under retention.

Bound	Rule
Artifact count/depth	policy budget; stop with explicit budget_exceeded
URL/redirects	deduplicate canonical identity while preserving observations
Media derivation	parent hash + deterministic/learned preprocessing provenance
Cross-tenant	never deduplicate in a way that leaks existence

5. Route planner and registry resolution

Aliases are presentation only. They cannot change scientific task, labels or deployment after resolution.

Plan deterministic and specialist work from validated capability and tenant policy. Before each HF dispatch, persist registry resolution with requested role, exact entry/deployment/revision/schema/calibration. Retry uses that deployment only. Shadow is a separate planned evidence item.

6. Evidence and inference receipts

Completed learned evidence requires a matching receipt. Deterministic evidence has code/config/version provenance but no fake model receipt. Unknown labels, non-finite scores, oversized responses or deployment mismatch fail.

Evidence	Fields
Required fields	evidence_id, investigation_id, artifact_id, specialist, scientific_role, registry_resolution_id, deployment_id, model_key, immutable_model_revision, preprocessing_version, input_hash, raw_output, raw_label, raw_score, normalized_label, normalized_score, calibrated_probability, calibration_version, quality, sufficiency, ood_status, status, reason_codes, started_at, completed_at
Lifecycle states (16)	planned, dispatched, running, completed, abstained, not_applicable, unavailable, timed_out, failed, cancelled, privacy_blocked, policy_skipped, insufficient, unsupported, degraded, budget_exceeded
Semantic rule	raw_score, normalized_score, and calibrated_probability are separate nullable fields and never silently substituted.

Receipt	Fields
inference_receipt.v2 fields	receipt_id, registry_resolution_id, deployment_id, deployment_bundle_id, request_id, provider, endpoint_identity, endpoint_deployment_revision, request_sha256, response_sha256, provider_request_id, components, attempt, latency_ms, billed_units, completion_status, received_at
components[] fields	component_role, model_key, repository_id, immutable_revision, artifact_or_container_digest, preprocessing_version, input_schema_version, output_schema_version, execution_order, qualification_status
Compatibility	A valid v1 receipt is exposed as a v2 receipt with one synthetic component derived from its singular repository/model/revision/preprocessing fields; missing historical provenance remains explicit and is never fabricated.

7. Processing modes, queues and deadlines

- Queues carry metadata/object references, not weights.
- Deadline and cancellation propagate to provider attempts.
- Idempotency covers create, upload completion and worker redelivery.

Mode	Contract	Queue/backpressure
Fast	complete inside caller budget	per-tenant admission; no long media
Hybrid	return partial/202	continue declared jobs
Async	poll/webhook	weighted fair workload queues; dead letter

8. Correlation, sufficiency, policy and review

Correlation is versioned deterministic reasoning over evidence, relationships and quality. It never treats missing as zero or averages incomparable specialist scores. Policy produces allow/review/hold/block/uncertain/unsupported/failed. Mandatory review triggers are explicit.

- High consequence + incomplete evidence.
- Deepfake in identity/financial context.
- Strong disagreement/OOD/provider degradation.
- Unsupported artifact/low calibration/critical action.

9. Database, API, events and webhooks

OpenAPI production host remains an implementation setting; the current example domain is a placeholder.

Preserve current scan/upload/report/list/cancel routes. Add proposed registry-resolution, evidence, review and privileged provider-health routes. Event/webhook envelopes carry tenant-safe IDs, status, policy/contract versions and HMAC signatures; payloads do not leak raw content by default.

10. Security and privacy

Preserve tenant HMAC, signed upload, idempotency, SSRF and webhook controls. Add endpoint/egress allowlists, scoped HF secrets, receipt identity validation, provider privacy profiles, log redaction and custom-handler/container supply-chain controls.

Threat	Control
Alias/deployment swap	pre-dispatch resolution + receipt match
SSRF/DNS rebinding	per-hop IP classification/pinning + egress
Webhook replay	HMAC timestamp nonce/idempotency
Cross-tenant leakage	RLS, tenant-scoped cache/dedupe, redacted telemetry
Provider compromise	schema bounds, circuit breaker, incident/disable pointer

11. Scaling, SLO discovery, cost and observability

Scale orchestration on queue age/deadlines and HF deployments on measured role needs. Use per-tenant concurrency, weighted fairness, dedupe and circuit breakers. Establish SLOs from load/cold-start tests rather than legacy estimates.

Metrics	Examples
Traffic	requests, bytes, artifact fan-out, queue age
Provider	latency, 429/5xx, cold starts, replica state
Contract	schema/label/receipt mismatch
Science	sufficiency, abstention, OOD, disagreement, drift
Cost	billed units, endpoint minutes, tenant/model budget

12. CLI, PowerShell and analyst experience

Verified browser CLI commands remain: help, clear, status, models, attach, detach, scan, link, phish, image, gateway, unified, history, open, version and exit. PowerShell setup/create/poll/upload/report/history routes match current API. New evidence/review commands are proposed and versioned.

- Never expose HF tokens/endpoints in client config.
- Show requested role, exact deployment display identity, status, missingness and receipt ID.
- Require reason/dual control for critical review actions.

13. Migration, first tickets and done

P0 containment, P1 contracts, P2 HF deployments, P3 qualification/shadow, P4 canary, P5 production, P6 contract/cleanup. First milestone is VT2-001 through VT2-016.

- Atomic registry/policy/calibration rollback.
- Provider outage drill and billing reconciliation.
- Done: no local learned inference, exact receipts, explicit failures, tenant isolation, current clients green.

References and package evidence

Primary sources and immutable model snapshots are listed in Research/Sources/source_ledger.csv and Research/Sources/HuggingFace/. Repository claims are traceable through repository_inventory.csv, source_file_audit.csv, runtime_call_graph.md, current_architecture_map.md and code_claim_ledger.csv.

- Architecture_Source_of_Truth.yaml - canonical invariants, roles, schemas and rollout.
- Model_Registry_Proposal.json - immutable role/deployment proposal.
- HuggingFace_Deployment_Matrix.csv - point-in-time provider and endpoint decisions.
- Technical_Findings_Register.csv / Implementation_Backlog.csv - evidence-backed remediation.
- Database_Target_Contract.md / Database_Migration_Specification.md - logical contract without fabricated SQL.
- Competitor_Analysis.md / Competitor_Matrix.csv - primary-source market design loop.

Authoritative external references

Hugging Face Inference Providers | Hugging Face Inference Endpoints | C2PA Technical Specification. Access dates, exact page URLs, claim summaries and verification status are preserved in the source ledger.

Universal operational controls

Security, privacy, cost, capacity, calibration, observability, incident response and rollback are release gates for every role. No UNKNOWN is replaced with an estimate; unresolved values remain explicit until an evidence-producing test or owner decision closes them.

Final boundary

Hugging Face-managed learned inference is authoritative. No learned production model runs inside the VeriTrust Vercel application or a VeriTrust-owned worker/server; no production weights are required in the VeriTrust deployment repository.